

Chapter 10 Mean Value theorem & maximum principle
for harmonic func.

u is harmonic $\Leftrightarrow -\Delta u = 0$.

< Mean Value theorem >

Assume $u: \mathbb{R} \rightarrow \mathbb{R}$

satisfies $-u''(x) = 0, \forall x \in \mathbb{R}$

Then $\frac{1}{2}(u(x) + u(-x))''$

$$= \frac{1}{2}(u''(x) + u''(-x))$$

$$= 0 \quad (\forall x \in \mathbb{R})$$

$$\Rightarrow \frac{1}{2}(u(x) + u(-x))' = \text{Const.}$$

$$\frac{1}{2}(u'(x) + u'(-x))$$

Plugging $x = 0$.

$$\Rightarrow \frac{1}{2}(u'(0) + u'(-0)) = \text{Const.}$$

$$\therefore \text{Const} = 0.$$

Plugging $x = 0$ again

$$\frac{1}{2}(u(0) + u(-0)) = \text{Const.}$$

$$\Rightarrow \text{Const} = u(0).$$

$$\therefore \forall r > 0 \quad (x = r).$$

$$\frac{1}{2}(u(r) + u(-r)) = u(0) \quad \leftarrow \text{mean of } u(r) + u(-r)$$

For then integrating $0 \leq r \leq R$.

$$\frac{1}{2} \int_0^R (u(r) + u(-r)) dr = R u(0)$$

$$\Leftrightarrow \frac{1}{2R} \int_{-R}^R u(r) dr = u(0)$$

↑
length of $(-R, R)$

↑ Integral mean Integral average

Thm 10.1 (Mean val thm)

Let $\Omega \subset \mathbb{R}^n$ be an open set, and let $u \in C^2(\Omega)$
be a harmonic func. ($-\Delta u = 0$).

Then for any $R > 0$ and $x \in \Omega$ satisfying $B_R(x) \subset \Omega$

$$u(x) = \frac{1}{|\partial B_R(x)|} \int_{\partial B_R(x)} u(y) d\sigma(y) \quad (10.3)$$

$$u(x) = \frac{1}{|B_R(x)|} \int_{B_R(x)} u(y) dy \quad (10.4)$$

pf. We show (10.3) first. For any $0 < r < R$.

$$\textcircled{\circ} = \int_{B_r(x)} \Delta u(y) dy = \int_{B_r(x)} \text{div}(\nabla u(y)) dy$$

↑
harmonic

$$= \int_{\partial B_r(x)} \nabla u(y) \cdot \nu(y) d\sigma(y)$$

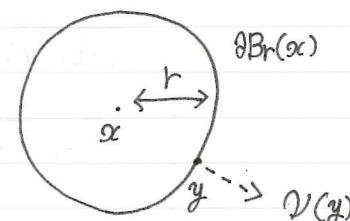
↑
div. thm

We introduce the polar coordinate.

$$y = r\omega + x, \quad r > 0, |\omega| = 1$$

↑ radius ↑ angle

$$\nu(y) = \frac{y - x}{r} = \frac{r\omega - x}{r} = \omega$$



and $d\tau(y) = r^{n-1} d\tau(\omega)$

$$\begin{aligned} 0 &= \int_{\partial B_r(x)} \nabla u(y) \cdot \nu(y) d\tau(y) \\ &= \int_{\partial B_r(x)} \nabla u(r\omega + x) \cdot \omega r^{n-1} d\tau(\omega) \\ &= \int_{\partial B_r(x)} \frac{\partial}{\partial r} u(r\omega + x) \cdot r^{n-1} d\tau(\omega) \\ &= r^{n-1} \frac{\partial}{\partial r} \int_{\partial B_r(x)} u(r\omega + x) d\tau(\omega) \\ &= r^{n-1} \frac{\partial}{\partial r} \left(\int_{\partial B_r(x)} u(y) \cdot r^{1-n} d\tau(y) \right) \end{aligned}$$

Therefore

$$\frac{\partial}{\partial r} \left(\int_{\partial B_r(x)} u(y) \cdot r^{1-n} d\tau(y) \right) = 0$$

$\Rightarrow$ There is a constant $C \in \mathbb{R}$ s.t

$$\int_{\partial B_r(x)} u(y) \cdot r^{1-n} d\tau(y) = C, \quad (0 < \forall r < R)$$

Note that $|\partial B_r(x)| = r^{n-1} |\partial B_1(x)|$.

Then

$$\frac{1}{|\partial B_r(x)|} \int_{\partial B_r(x)} u(y) d\tau(y) = \frac{C}{|\partial B_r(x)|}$$

Now taking a limit $r \downarrow 0$.

$$\frac{1}{|\partial B_r(x)|} \int_{\partial B_r(x)} u(y) d\tau(y) \rightarrow u(x)$$

$$\therefore \frac{C}{|\partial B_r(x)|} = u(x)$$

Therefore

$$\frac{1}{|\partial B_r(x)|} \int_{\partial B_r(x)} u(y) d\tau(y) = u(x), \quad (0 < \forall r < R)$$

Next we show (10.4)

$$\begin{aligned} \int_{\partial B_R(x)} u(y) dy &= \int_0^R \left(\int_{\partial B_r(x)} u(y) d\tau(y) \right) dr \\ &\stackrel{\text{polar}}{=} \int_0^R u(x) |\partial B_r(x)| dr \\ &= u(x) \int_0^R r^{n-1} |\partial B_r(x)| dr \\ &= u(x) \frac{1}{n} R^n |\partial B_r(x)| \\ &\quad \quad \quad |\partial B_r(x)| \\ &= u(x) \cdot |B_R(x)| \end{aligned}$$

$$\Rightarrow \frac{1}{|B_r(x)|} \int_{B_r(x)} u(y) dy = u(x).$$

Note

	dim	Volume	Surface area
radius: r	1	$2r$	2
$B_r(0)$	2	πr^2	$2\pi r$
	3	$\frac{4}{3}\pi r^3$	$4\pi r^2$
	4	$\frac{1}{4} \partial B_r(0) r^4$	$ \partial B_r(0) r^3$

<Maximum Principle>

Let $u \in C^2(-1, 1) \cap C([-1, 1])$

be a harmonic func. ($-u'' = 0$)

Then there is a constant $C \in \mathbb{R}$ s.t. $-u'(x) = C$

$$\Rightarrow u(x) = -C_1 x + C_2$$

If $C_1 > 0$ then $u(-1) \geq u(x) \geq u(1)$, $\forall x \in [-1, 1]$

If $C_1 < 0$ then $u(-1) \leq u(x) \leq u(1)$, $\forall x \in [-1, 1]$

If $C_1 = 0$ then $u(-1) \leq u(x) \leq u(1)$, $\forall x \in [-1, 1]$

↑ equal ↓

Thm 10.2 (Maximum Principle)

Let $\Omega \subset \mathbb{R}^n$ be a domain and Let $u \in C^2(\Omega) \cap C(\bar{\Omega})$

be a harmonic func. ($-\Delta u = 0$)

Then $\max_{x \in \bar{\Omega}} u(x) = \max_{x \in \partial \Omega} u(x)$ holds (weak maximum principle)

Furthermore $\exists x_0 \in \Omega$ s.t. $\max_{x \in \bar{\Omega}} u(x) = u(x_0)$

$\Rightarrow u$ is constant.

holds (strong maximum principle).

pf. We show strong maximum principle

Assume that $\exists x_0 \in \Omega$ s.t. $\max_{x \in \bar{\Omega}} u(x) = u(x_0) =: M$

Take $B_r(x_0) \subset \Omega$ ($\because \Omega$ is open)

By mean val thm.

$$M = u(x_0) = \frac{1}{|B_r(x_0)|} \int_{B_r(x_0)} u(y) dy$$

$$\Leftrightarrow \frac{1}{|B_r(x_0)|} \int_{B_r(x_0)} (M - u(y)) dy = 0$$

$$\Rightarrow M - u(y) = 0 \text{ in } B_r(x_0)$$

Therefore $\{y \in \Omega; u(y) = M\} = u^{-1}(\{M\})$
is an open set

Since u is continuous on $\bar{\Omega}$

$$u^{-1}(\{M\}) = \{y \in \Omega; u(y) = M\}$$

is a closed set

$\therefore \{y \in \Omega; u(y) = M\}$ is open and closed (clopen)

Since Ω is connect clopen sets are only empty sets or the whole set Ω

$$\Rightarrow \{y \in \Omega; u(y) = M\} = \Omega$$

$$\Rightarrow u(y) = M \quad (\forall y \in \Omega)$$

$\therefore$ strong maximum principle holds.

Next $x_0 \in \bar{\Omega}$ satisfy $u(x_0) = \max_{x \in \bar{\Omega}} u(x)$

we show weak maximum principle Let

If $x_0 \in \partial \Omega \Rightarrow u(x_0) = \max_{x \in \partial \Omega} u(x)$

If $x_0 \in \Omega \Rightarrow u$ is constant in Ω by strong max principle

$$\Rightarrow \max_{x \in \bar{\Omega}} u(x) = \max_{x \in \partial \Omega} u(x) \text{ by } u \in C(\bar{\Omega}) \quad \square$$

Remark: If $u \in C^2(\Omega) \cap C(\bar{\Omega})$ is harmonic func.

$$\Rightarrow \max_{x \in \bar{\Omega}} (-u(x)) = \max_{x \in \partial \Omega} (-u(x))$$

$$\Rightarrow -\min_{x \in \bar{\Omega}} (u(x)) = -\min_{x \in \partial \Omega} (u(x))$$

$$\Rightarrow \min_{x \in \bar{\Omega}} (u(x)) = \min_{x \in \partial \Omega} (u(x))$$

Thm 10.3

Let $\Omega \in \mathbb{R}^n$ be a domain, let f be a continuous func. g be a continuous func. on $\partial\Omega$ Assume $u, v \in C^2(\Omega) \cap C(\overline{\Omega})$ are satisfy
sol. of

$$\begin{cases} -\Delta u(x) = f(x) ; x \in \Omega \\ u(x) = g(x) ; x \in \partial\Omega \end{cases}$$

$$\begin{cases} -\Delta v(x) = f(x) ; x \in \Omega \\ v(x) = g(x) ; x \in \partial\Omega \end{cases}$$

then $u = v$ in Ω pf. let $w = u - v$

$$\Rightarrow \begin{cases} -\Delta(u-v)(x) = 0 ; x \in \Omega \\ (u-v)(x) = 0 ; x \in \partial\Omega \end{cases}$$

$$\Rightarrow \begin{cases} -\Delta w(x) = 0 ; x \in \Omega \\ w(x) = 0 ; x \in \partial\Omega \end{cases}$$

By weak maximum principle

$$\min_{x \in \partial\Omega} w(x) \leq w(x) \leq \max_{x \in \partial\Omega} w(x) ; \forall x \in \Omega$$

$$0 \leq w(x) \leq 0 \Rightarrow w(x) = 0 ; \forall x \in \Omega$$

$$\therefore u(x) - v(x) = 0.$$

$$\Leftrightarrow u(x) = v(x) ; \forall x \in \Omega \quad \square$$